

Markov's Inequality and Stochastic Ordering

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Abstract

Markov's inequality is a foundational tool in probability. Here, I describe a simple re-interpretation of the content of Markov's inequality, and detail some simple extensions.

Recalling Markov's Inequality

Let X be a non-negative random variable of finite expectation. Markov's inequality gives that

$$t \geq 0 \implies \mathbf{P}(X \geq t) \leq \min \left\{ 1, \frac{\mathbf{E}[X]}{t} \right\}.$$

Equally, one can reparameterise this in the form

$$\alpha \in [0, 1] \implies \mathbf{P}\left(X \geq \frac{\mathbf{E}[X]}{\alpha}\right) \leq \alpha.$$

One can recognise the expression on the right-hand side as the cumulative distribution function of random variable U drawn uniformly on the interval $[0, 1]$, i.e.

$$\alpha \in [0, 1] \implies \mathbf{P}\left(X \geq \frac{\mathbf{E}[X]}{\alpha}\right) \leq \mathbf{P}(U \leq \alpha).$$

A simple re-writing transforms this into

$$\alpha \in [0, 1] \implies \mathbf{P}\left(\frac{\mathbf{E}[X]}{X} \leq \alpha\right) \leq \mathbf{P}(U \leq \alpha),$$

or

$$\alpha \in [0, 1] \implies \mathbf{P}(U > \alpha) \leq \mathbf{P}\left(\frac{\mathbf{E}[X]}{X} > \alpha\right).$$

We will see that this elucidates a new interpretation of Markov's inequality in terms of stochastic ordering.

Stochastic Ordering

Say that two $\mathbf{R}$ -valued random variables X and Y are ordered in the sense of stochastic ordering (write $X \preceq_{\text{st}} Y$) if for all increasing functions F , there holds the inequality

$$\mathbf{E}[F(X)] \leq \mathbf{E}[F(Y)],$$

or equivalently, if for all $t \in \mathbf{R}$, there holds the inequality

$$\mathbf{P}(X \geq t) \leq \mathbf{P}(Y \geq t).$$

This codifies the idea that ' X is smaller than Y ' in some universal sense.

Accepting this definition, we can interpret Markov's inequality as the statement that

$$X \geq 0, \quad \mathbf{E}[X] < \infty \implies U \preceq_{\text{st}} \frac{\mathbf{E}[X]}{X},$$

or equivalently,

$$X \geq 0, \quad \mathbf{E}[X] < \infty \implies X \preceq_{\text{st}} \frac{\mathbf{E}[X]}{U},$$

noting that the stochastic ordering behaves in simple ways under composition with monotone functions.

Composition

Extending this reasoning further, let T be a strictly increasing and non-negative transformation such that $\mathbf{E}[T(X)] < \infty$. Applying the preceding reasoning to $X' = T(X)$, see that

$$T(X) \preceq_{\text{st}} \frac{\mathbf{E}[T(X)]}{U}.$$

Applying T^{-1} on both sides, we conclude that

$$X \preceq_{\text{st}} T^{-1}\left(\frac{\mathbf{E}[T(X)]}{U}\right).$$

One can imagine taking $T(x) = x^p$, $T(x) = \exp(\lambda \cdot x)$, etc. to obtain concrete implications, depending on the application of interest.

Extension

If we know more about X , we can obtain similar dominations, though with a somewhat messier interpretation. For example, given $X \geq 0$ with a known mean and variance, one can compute the bound

$$t \geq 0 \implies \mathbf{P}(X \geq t) \leq \min \left\{ 1, \frac{\mathbf{E}[X]}{t}, \frac{\mathbf{Var}[X]}{\mathbf{Var}[X] + (t - \mathbf{E}[X])^2} \right\},$$

with behaviours split across $t \in [0, \mathbf{E}[X]]$, $t \in \left[\mathbf{E}[X], \frac{\mathbf{E}[X^2]}{\mathbf{E}[X]}\right]$, and $t \geq \frac{\mathbf{E}[X^2]}{\mathbf{E}[X]}$. This can be seen as a hybrid of the usual Markov inequality with Cantelli's inequality.

The right-hand side is again interpretable as a distribution function. Define $\bar{X}$ to have density given by

$$f_{\bar{X}}(x) = \begin{cases} 0 & 0 < x < \mathbf{E}[X] \\ \frac{\mathbf{E}[X]}{x^2} & \mathbf{E}[X] < x < \frac{\mathbf{E}[X^2]}{\mathbf{E}[X]} \\ \frac{2 \cdot \mathbf{Var}[X] \cdot (x - \mathbf{E}[X])}{(\mathbf{Var}[X] + (x - \mathbf{E}[X])^2)^2} & \frac{\mathbf{E}[X^2]}{\mathbf{E}[X]} < x, \end{cases}$$

noting its discontinuities at $x \in \left\{\mathbf{E}[X], \frac{\mathbf{E}[X^2]}{\mathbf{E}[X]}\right\}$. One then infers that $X \preceq_{\text{st}} \bar{X}$, though it is not obvious that one can obtain a simplified characterisation of $\bar{X}$ (as e.g. a more straightforward function of $\mathbf{E}[X]$, $\mathbf{E}[X^2]$, and a uniform random variable).

Conclusion

Markov's inequality turns a bound on an expectation into a bound on tail probabilities, and this logically entails a statement about stochastic ordering. This can be extended painlessly in certain ways, though this extension is a bit more challenging to convey concisely when one has bounds on multiple distinct expectations.